

GrounUp Enterprise Specification

Phase 26 - Estimating, Cost, Production, Crew/Equipment, Haul/Disposal, Markup & Execution Engine

Version 1.0.0 | Implementation-ready specification

1. Mission

Implement the deterministic calculation and lifecycle engine that converts scope, quantities and Phase 25 governed library data into reproducible direct cost, production, duration, hauling, markup and sell-price outputs.

2. Phase 25 Dependency

Every authoritative estimate uses an immutable Phase 25 snapshot or explicit immutable record versions for service/resource/rate inputs. This prevents historical estimates from changing when live libraries change.

3. Production, Crew and Duration

Production drives duration; duration drives crew labor and participating equipment runtime. Crew-size and equipment alternatives can be compared without losing source assumptions.

4. Equipment and Fuel

Equipment rate selection uses governed hourly/daily/weekly/monthly policy. Fuel consumption is calculated from runtime and configured consumption/load assumptions.

5. Trucking, Haul and Disposal

The engine models load, outbound, dump/queue, return and delay time; trips, truck count and hourly/trip cost are explicit. Disposal uses governed quantity/tonnage conversions and fees.

6. Cost Rollup

Labor, equipment, fuel, material, subcontract, trucking, disposal, mobilization and other governed direct costs roll into deterministic direct cost.

7. Markup and Sell Price

Indirects, contingency, overhead, profit, tax and bond follow configured calculation bases and order. Parallel and stacked methods are not silently interchanged.

8. Scenarios and Confidence

Low/base/high and alternate methods preserve explicit assumption sets. Confidence reflects controlled evidence/completeness/review factors rather than hidden AI judgment.

9. AI Estimate Assist

AI may suggest mappings, assumptions, risks, crews and equipment alternatives with evidence/confidence, but authoritative values change only through explicit user/governed actions.

10. Writeback and Handoff

Approved new services/assemblies can flow to Phase 25 through governed writeback. Approved/frozen estimate versions export exact baseline/version/snapshot data to proposals and job costing.

Implementation Modules

ID	Module	Layer	Purpose
P26-MOD-001	Estimate Service	Domain	Estimate header, lifecycle, versions, ownership, status and approval
P26-MOD-002	Estimate Line Service	Domain	Service/line-item selection, quantities, units, notes and source evidence
P26-MOD-003	Assembly Expansion Engine	Engine	Expand versioned assemblies into estimate resource components
P26-MOD-004	Library Snapshot Resolver	Engine	Resolve immutable Phase 25 library snapshot/version inputs
P26-MOD-005	Quantity Engine	Engine	Normalize input/takeoff quantities and approved unit conversions
P26-MOD-006	Production Engine	Engine	Determine base production and modifiers for task/scenario
P26-MOD-007	Crew Engine	Engine	Resolve crew composition, crew size, labor hours and crew alternatives
P26-MOD-008	Labor Cost Engine	Engine	Base labor + burden + overtime/shift rules where configured
P26-MOD-009	Equipment Selection Engine	Engine	Select compatible equipment class/model options for work
P26-MOD-010	Equipment Cost Engine	Engine	Apply hourly/daily/weekly/monthly rate policy to required duration
P26-MOD-011	Fuel Engine	Engine	Estimate fuel quantity/cost from configured consumption and runtime assumptions
P26-MOD-012	Truck Cycle Engine	Engine	Load/haul/dump/return cycle model with capacity and delays
P26-MOD-013	Truck Count Engine	Engine	Determine truck count required to balance source/placement production
P26-MOD-014	Haul Cost Engine	Engine	Trip/hour/load-based hauling cost with distance and cycle evidence
P26-MOD-015	Disposal Engine	Engine	Disposal quantity/fee/tonnage and facility/source logic
P26-MOD-016	Material Cost Engine	Engine	Material quantity, waste, price, freight/source and unit conversion
P26-MOD-017	Subcontract Cost Engine	Engine	Subcontract quantity/rate/quote resolution
P26-MOD-018	Duration Engine	Engine	Work quantity divided by governing production adjusted for shifts/calendars
P26-MOD-019	Mobilization Engine	Engine	Equipment/crew mobilization and demobilization cost/time
P26-MOD-020	Indirect Cost Engine	Engine	Project-specific indirect/general-condition allocation
P26-MOD-021	Regionalization Engine	Engine	Apply allowed regional/location profile factors without double adjustment
P26-MOD-022	Markup Engine	Engine	Configured markup sequence, overhead, profit, contingency and exclusions
P26-MOD-023	Tax & Bond Engine	Engine	Configured tax/bond treatment and calculation basis
P26-MOD-024	Unit Price Engine	Engine	Calculate sell unit price from total sell amount and quantity
P26-MOD-025	Scenario Engine	Engine	Low/base/high and alternate crew/equipment/haul/shift comparisons
P26-MOD-026	Confidence Engine	Engine	Input completeness/source age/override/review confidence scoring
P26-MOD-027	AI Estimate Assist	AI	Suggest mappings/assumptions/risks with evidence; never silently changes approved
P26-MOD-028	Estimate Review Workflow	Governance	Review, approval, exceptions and estimate freeze
P26-MOD-029	Service Catalog Writeback	Integration	Approved new service/assembly definitions written to Phase 25 catalog
P26-MOD-030	Estimate Baseline Export	Integration	Proposal/job-cost/schedule/project handoff with immutable estimate baseline

Core Formula Registry

P26-FRM-001 - Duration Hours: $\text{quantity} / \text{production_rate}$ | quantity and production rate must have compatible output units

P26-FRM-002 - Crew Labor Hours: $\text{duration_hours} * \text{crew_member_count}$ | sum across crew roles

P26-FRM-003 - Labor Cost: $\text{labor_hours} * \text{loaded_labor_rate}$ | or explicit base + burden components

P26-FRM-004 - Equipment Cost: $\text{selected_rate_policy}(\text{runtime, hourly, daily, weekly, monthly})$ | rate policy governs cheapest/contracted/actual rental basis

P26-FRM-005 - Fuel Quantity: $\text{runtime_hours} * \text{fuel_consumption_per_hour} * \text{load_factor}$ | load_factor optional/default per policy

P26-FRM-006 - Fuel Cost: $\text{fuel_quantity} * \text{fuel_unit_price}$ | effective-dated fuel price

P26-FRM-007 - Truck Cycle Minutes: $\text{load} + \text{outbound} + \text{dump_queue} + \text{return} + \text{delays}$ | all components explicit

P26-FRM-008 - Truck Trips: $\text{ceil}(\text{haul_quantity} / \text{effective_payload})$ | unless partial-load policy differs

P26-FRM-009 - Truck Count: $\text{ceil}(\text{source_production_per_hour} * \text{cycle_hours} / \text{payload})$ | balance rule may use target utilization/buffer

P26-FRM-010 - Hourly Haul Cost: $\text{truck_count} * \text{truck_hourly_rate} * \text{haul_duration_hours}$ | plus configured mobilization/fees

P26-FRM-011 - Trip Haul Cost: $\text{trips} * \text{trip_rate}$ | trip rate may derive from cycle and hourly rate

P26-FRM-012 - Disposal Tons: $\text{volume} * \text{approved_density_conversion}$ | only when density is governed

P26-FRM-013 - Material Purchase Qty: $\text{installed_qty} * (1 + \text{waste_factor})$ | waste factor decimal

P26-FRM-014 - Material Cost: $\text{purchase_qty} * \text{unit_price} + \text{configured freight}$ | regional/vendor rule applies

P26-FRM-015 - Direct Cost: sum governed direct cost components | component categories explicit

P26-FRM-016 - Sell Price: apply configured markup/contingency/O&P;/tax/bond sequence to governed bases | parallel vs stacked preserved

P26-FRM-017 - Unit Sell Price: $\text{sell_price} / \text{priced_quantity}$ | quantity > 0

P26-FRM-018 - Production Adjusted: $\text{base_production} * \text{product_of_applicable_rate_modifiers}$ | if modifier policy is production-based

P26-FRM-019 - Duration Adjusted: $\text{base_duration} * \text{product_of_applicable_duration_modifiers}$ | if modifier policy is duration-based

P26-FRM-020 - Confidence: weighted score of input/source/review factors | weights/version governed